

Skills Summary

Dividing by a One-Digit Number

Use this reference while completing your worksheet or when studying.

Skill 1 — Long Division, Digit by Digit

Four moves, repeated: divide, multiply, subtract, bring down.

Work left to right through the dividend. Each answer digit goes *directly above* the digit you just used, so the places stay lined up.

If the divisor does not go into the digit you have, write 0 above it and bring the next digit down. Stop when there is nothing left to bring down; whatever is left below is the remainder, and it is always smaller than the divisor.

Example: $738 \div 5$

Step 1. The divisor is bigger than the first digit is comfortable with, so I work through the dividend one digit at a time and keep each answer digit above the digit it came from.

Step 2. 5 into 7 goes once, leaving 2; bring down 3 to make 23, and 5 goes 4 times leaving 3; bring down 8 to make 38, and 5 goes 7 times leaving 3.

$$738 \div 5 = 147 \text{ remainder } 3 \quad (\text{check: } 147 \times 5 + 3 = 738)$$

Try it: $926 \div 4$. What is the quotient?

check: 231 remainder 2

Skill 2 — Partial Quotients

Take out easy chunks of groups, then add the chunks.

Pick a friendly multiple of the divisor — 10, 50, 100, 600 groups — subtract it, and repeat on what is left.

Any chunks work; smaller chunks just take more rounds. The chunk sizes you took out are the partial quotients, and their sum is the quotient.

Example: $336 \div 3$ with partial quotients

Step 1. I would rather subtract friendly chunks than juggle carries, so I take out a round number of groups first and finish with what is left.

Step 2. $3 \times 100 = 300$, and $336 - 300 = 36$; then $36 \div 3 = 12$, so the chunks are 100 and 12.

$$336 \div 3 = 100 + 12 = 112$$

Try it: $856 \div 8$ with partial quotients (start with 100 groups of 8).

check: 107

Skill 3 — Estimate First, Check After

Before: round the dividend to a friendly number and divide that. The estimate tells you how many digits the quotient should have and roughly how big it is.

After: multiply back. $(\text{quotient}) \times (\text{divisor}) + (\text{remainder})$ must rebuild the dividend exactly.

Rounding the dividend *up* makes the estimate a little large; rounding *down* makes it a little small.

Example: Round 2,874 to the nearest thousand, ready to estimate $2,874 \div 6$.

Step 1. I round only the dividend, because a friendly dividend is what makes the estimate easy while the divisor stays exact.

Step 2. Rounding to the thousand means looking at the hundreds digit, 8; that is 5 or more, so the number rounds up.

2,874 to the nearest thousand is **3000**, so the estimate is $3000 \div 6 = 500$

Try it: Round 5,390 to the nearest thousand, ready to estimate a division.

check: 0005

(!) Watch out: a 0 in the quotient is a digit, not a gap. In $618 \div 6 = 103$ the zero holds the tens place; leaving it out gives 13, which the multiply-back check catches at once.